X-linked Cleft Palate With or Without Ankyloglossia (CPX) — Comprehensive Disease Report

Disease: Cleft Palate With Or Without Ankyloglossia, X-linked (CPX / CPA) **Causal gene: TBX22** (T-box transcription factor 22) **MONDO:** MONDO:0010560 · **OMIM:** 303400 · **Orphanet:** 324601 · **ICD-10:** Q35 (cleft palate) · **MeSH:** C536426 · **DOID:** 0060613 · **GARD:** 0001394 · **UMLS:** C1844830 · **MedGen:** 375520 · **SNOMED CT:** 766761000 **Category:** Mendelian (X-linked semidominant)

> **Gene statement (as required):** The disease described here, *X-linked cleft palate with or without ankyloglossia* (CPX; also "cleft palate with ankyloglossia," CPA), is caused by loss-of-function mutations in **TBX22**, the gene encoding the **T-box transcription factor 22**, located at **Xq21.1**. This is distinct from autosomal forms of ankyloglossia (e.g., the autosomal-dominant ankyloglossia with tooth anomalies family, PMID: 29704302), which are **not** TBX22-related.

Summary

X-linked cleft palate with or without ankyloglossia (**CPX**, OMIM 303400, MONDO:0010560) is a Mendelian craniofacial malformation caused by **loss-of-function mutations in TBX22**, a gene at **Xq21.1** encoding a SUMOylation-dependent **T-box transcription factor** that acts as a **transcriptional repressor**. The condition is defined clinically by the combination of **cleft palate** (complete, partial, or submucous) and **ankyloglossia** (tongue-tie from a short lingual frenulum), frequently accompanied by a **bifid, absent, or shortened uvula**. It was first mapped to Xq21.1 by linkage in large Icelandic, British Columbia Native, and Manitoba Mennonite kindreds PMID: 8045560, and the causal gene was identified by Braybrook and colleagues in 2001 PMID: 11559848.

The molecular mechanism is a **complete loss of TBX22 function**. Truncating mutations (nonsense, frameshift, splice-site) and T-box-domain missense mutations abolish the protein's ability to bind DNA and/or to be modified by **SUMO-1 at lysine 63**, which is required for its

repressor activity PMID: 17846996. Developmentally, the *Tbx22*-null mouse demonstrates that the gene is required for **osteogenic patterning of the posterior hard palate**: ossification is severely reduced after condensation of the palatal mesenchyme due to a **delay in osteoblast maturation**, producing a submucous cleft rather than a failure of palatal shelf closure, together with ankyloglossia and choanal atresia PMID: 19648291. TBX22 expression in the palatal shelves, the base of the tongue at the frenulum, and the nasal septum precisely mirrors the human phenotype PMID: 12374769.

CPX is inherited as a **semidominant X-linked trait**: hemizygous males are typically more severely affected (overt or submucous cleft palate plus ankyloglossia), while heterozygous carrier females show variable, usually milder manifestations (ankyloglossia, bifid uvula, submucous cleft) consistent with X-inactivation. The phenotypic spectrum extends from **isolated ankyloglossia** through classic CPX to **CHARGE-like Abruzzo-Erickson syndrome** (ABERS) with coloboma and sensorineural deafness. Management follows the standard multidisciplinary cleft care pathway — surgical palatoplasty, frenuloplasty for tongue-tie, and speech, hearing, and dental therapy — combined with X-linked genetic counseling. No gene-targeted therapy exists.

Key Findings

Finding 1 — TBX22 is the causal gene for CPX

CPX was localized by linkage analysis to **Xq21.1** (originally Xq21, with a multipoint Z = 13.5 near marker DXS447) in Manitoba Mennonite and British Columbia Native kindreds; "the most likely location for CPX is at DXS447 in Xq21.1" PMID: 8045560. In 2001, Braybrook et al. identified the causal gene, showing that "**CPX is caused by mutations in the gene encoding the recently described T-box transcription factor TBX22**" and that "the spectrum of nonsense, splice-site, frameshift and missense mutations we have identified in this study indicates that the **cleft phenotype results from a complete loss of TBX22 function**" PMID: 11559848. This establishes both the gene and its loss-of-function disease mechanism.

Verified gene identifiers: HGNC:11600 · NCBI Gene 50945 · UniProt Q9Y458 · Ensembl ENSG00000122145 · cytoband Xq21.1 · GRCh38 chrX:80,014,747–80,031,908 · gene MIM 300307.

Finding 2 — Missense mutations impair DNA binding, SUMOylation, and transcriptional repression

Andreou et al. (2007) defined the biochemical mechanism. TBX22 recognizes a palindromic DNA sequence based on **AGGTGTGA**, functions as a **transcriptional repressor**, and is a target for **SUMO-1 modification at lysine 63** (immediately upstream of the T-box domain). "We find that TBX22 is a target for the small ubiquitin-like modifier SUMO-1 and that **this modification is required for TBX22 repressor activity**" and "**loss of SUMO-1 modification is consistently found in all pathogenic CPX missense mutations**. This implies a general mechanism linking the loss of SUMO conjugation to the loss of TBX22 function" PMID: 17846996. All ten naturally occurring missense mutations examined — located within the T-box DNA-binding domain — impaired repression: "**TBX22 missense mutations result in impaired repression activity**."

A genotype–function correlation was further refined by Kantaputra et al. (2011): T-box-domain mutations **R120Q, R126W, and R151L** affect DNA binding and/or transcriptional repression, whereas the C-terminal mutations **P389Q and S400Y did not affect TBX22 activity**, indicating that pathogenicity maps to the T-box domain PMID: 21248356.

Finding 3 — The Tbx22-null mouse recapitulates CPX via reduced posterior palatal ossification

Pauws et al. (2009) generated a *Tbx22*-null mouse that "has a **submucous cleft palate (SMCP) and ankyloglossia, similar to the human phenotype**, with a small minority showing overt clefts." Persistent oro-nasal membranes produced **choanal atresia** and roughly **50% postnatal lethality** from breathing and feeding difficulty. The developmental mechanism is osteogenic: "**Tbx22 plays an important role in the osteogenic patterning of the posterior hard palate. Ossification is severely reduced after condensation of the palatal mesenchyme, resulting from a delay in the maturation of osteoblasts**" PMID: 19648291. Crucially, the defect is **not** a failure of palatal shelf elevation or closure — the shelves meet — but a deficit of intramembranous bone in the posterior hard palate producing the classic submucous notch.

Expression studies anchor the mechanism anatomically: TBX22 mRNA "is also detected in the **base of the tongue in the region of the frenulum that corresponds to the ankyloglossia** seen in CPX patients," and is expressed in the palatal shelves (highest before elevation), nasal septum, and tooth buds PMID: 12374769.

Finding 4 — Phenotypic spectrum from isolated ankyloglossia to Abruzzo-Erickson syndrome

TBX22-related disease spans a continuum. At the mild end, isolated ankyloglossia and dental anomalies occur; Kantaputra et al. expanded the spectrum "to include **dental anomalies and cleft lip**," with micro/hypodontia and cleft lip ± palate in some families PMID: 21248356. Demeer et al. (2018) documented "orofacial features previously not associated with the CPX phenotypic spectrum: **choanal atresia, Pierre-Robin sequence, and overgrowths on the posterior edge of the hard palate**" PMID: 30462376. At the severe end, TBX22 splice mutations cause **X-linked CHARGE-like Abruzzo-Erickson syndrome (ABERS)** with coloboma and sensorineural deafness: this work "confirms that **TBX22 is the gene underlying Abruzzo-Erickson syndrome, expanding the phenotypic spectrum of TBX22 mutations**" PMID: 22784330; a 2026 case series proposes that gain-of-function alleles may underlie ABERS specifically PMID: 41630180.

Upstream regulation of *Tbx22* is by opposing craniofacial signals: "**FGF8 induces Tbx22 in the early face while BMP4 represses and thus restricts its expression**," a mechanism conserved between chick and mouse PMID: 20845426. TBX22 mutations account for approximately **5% of non-syndromic cleft palate** patients.

Finding 5 — Gene–environment interaction: methylation and promoter variants

Two lines of evidence connect TBX22 to environmentally modulated and multifactorial clefting. In an **all-trans retinoic acid (ATRA)-induced** mouse cleft palate model, "**TBX22 exon 5 was hyper-methylated at the CpG site at E13.5 (P=0.025, log2FC=1.5) and E14.5 (P=0.011, log2FC=1.5)**" in treated embryos, with TBX22 expression negatively correlated with methylation and normalization by E16.5 (P=0.808) — linking a teratogen to epigenetic downregulation of TBX22 PMID: 31241704. In humans, two TBX22 promoter SNPs were associated with non-syndromic cleft lip ± palate (NSCLP) risk in a sex-specific manner: "**Significant associations of rs7055763 (P=0.034) and rs41307258 (P=0.022) with NSCLP were found only in women**," consistent with X-linked dosage sensitivity, with strong linkage disequilibrium (D'=0.97, r^2=0.77) PMID: 25918826.

Finding 6 — Semidominant X-linked inheritance with carrier-female manifestation

CPX is inherited as a **semidominant X-linked** disorder. In the British Columbia Native kindred, "**ankyloglossia (tongue-tie) is also a feature of X-linked cleft palate in some of the affected males and carrier females**" PMID: 1570839, establishing that heterozygous carriers can manifest. Hemizygous males are typically more severely affected (overt or submucous cleft palate plus ankyloglossia); carrier females show variable, usually milder features (ankyloglossia, bifid uvula, submucous cleft) consistent with random X-inactivation.

Finding 7 — TBX22 is strongly loss-of-function intolerant in gnomAD

Population constraint data corroborate the loss-of-function mechanism. For ENSG00000122145 (GRCh38), gnomAD reports **pLI = 0.990**, observed/expected LoF = 0.27 with **LOEUF = 0.48** (90% CI 0.16–0.48), **8 observed vs 30.1 expected** LoF variants, and **LoF Z = 3.42** — marked depletion of loss-of-function variants in the general population. By contrast, missense is **not** constrained (mis_z = −1.33; oe_mis = 1.16), consistent with pathogenicity being restricted to specific T-box-domain missense alleles rather than general missense intolerance.

Finding 8 — ClinVar variant landscape

A ClinVar query for `TBX22[gene]` returned **327 variation records**: **~170 Pathogenic, 13 Likely pathogenic, 161 Uncertain significance (VUS), and 122 Benign/Likely benign**. The CPX-causing intragenic set comprises nonsense, frameshift, splice-site, and T-box missense variants; the high Pathogenic count is partly attributable to large overlapping copy-number and contiguous-gene deletions at Xq21 rather than CPX point mutations alone.

Finding 9 — Core HPO phenotype annotations

The Monarch Initiative (v3 API) returns exactly three formal phenotype annotations for MONDO:0010560:

HPO ID	Term
HP:0010296	Ankyloglossia
HP:0000193	Bifid uvula
HP:0000175	Cleft palate

Expanded-spectrum HPO terms confirmed in the literature include **HP:0000176** (submucous cleft of the hard palate), **HP:0000185** (cleft soft palate), **HP:0000204** (cleft upper lip), **HP:0000668** (hypodontia), **HP:0000453** (choanal atresia), **HP:0000201** (Pierre-Robin sequence), **HP:0000589** (coloboma), and **HP:0000407** (sensorineural hearing impairment). (Note: HP:0000190 is "Abnormal oral frenulum morphology," distinct from bifid uvula HP:0000193.)

Finding 10 — Verified cross-references and Orphanet description

Monarch cross-references for MONDO:0010560: OMIM:303400, Orphanet:324601, MeSH:C536426, DOID:0060613, GARD:0001394, MedGen:375520, UMLS:C1844830, SNOMED CT:766761000. The Orphanet definition: the disease is "characterized by the association of complete, partial or submucous cleft palate and ankyloglossia. Patients may also present abnormal uvula (e.g. absent, bifid, shortened or laterally deviated), short lingual frenulum and dental anomalies... **Digital abnormalities, such as mild clinodactyly and/or syndactyly, have also been reported**."

Section-by-Section Report

1. Disease Information

CPX is a congenital craniofacial malformation defined by the co-occurrence of **cleft palate** (complete, partial, or submucous) and **ankyloglossia** (tongue-tie). It is a Mendelian, X-linked semidominant condition caused by TBX22. Key identifiers are listed in the header. Common synonyms: *cleft palate with ankyloglossia*; *X-linked cleft palate and ankyloglossia*; *CPX*; *CPA*. Information in this report derives from **aggregated disease-level resources** (OMIM, Orphanet, Monarch, ClinVar, gnomAD) and primary literature (family linkage studies, functional assays, mouse models), not from individual EHR records.

2. Etiology

Causal factor: germline loss-of-function mutations in **TBX22** (Xq21.1). The disease mechanism is a **complete loss of TBX22 function** PMID: 11559848.

Genetic risk factors: being hemizygous for a pathogenic TBX22 allele (males) confers near-complete risk of the malformation; carrier females are at risk of milder manifestations. Common promoter SNPs **rs7055763** and **rs41307258** modify non-syndromic cleft risk in women PMID: 25918826.

Environmental risk factors: general orofacial-cleft risk factors (maternal smoking, alcohol, folate deficiency, retinoid exposure) apply to clefting broadly; a retinoic-acid teratogen model specifically downregulates TBX22 via exon-5 hypermethylation PMID: 31241704.

Protective factors: no TBX22-specific protective alleles are established. Periconceptional folate is protective for orofacial clefts generally (not CPX-specific; evidence not derived in this investigation).

Gene–environment interaction: teratogen-induced hypermethylation of TBX22 provides a direct GxE link between retinoid exposure and reduced TBX22 dosage PMID: 31241704.

3. Phenotypes

Phenotype	HPO	Type	Onset	Frequency / notes
Cleft palate (overt/ submucous)	HP:0000175 / HP:0000176	Physical manifestation	Congenital	Core feature; males often overt, SMCP common
Ankyloglossia	HP:0010296	Physical manifestation	Congenital	Core; present in males and carrier females
Bifid/absent/short uvula	HP:0000193	Clinical sign	Congenital	Frequent
Hypodontia / dental anomalies	HP:0000668	Physical manifestation	Childhood	Expanded spectrum
Cleft lip ± palate	HP:0000204	Physical manifestation	Congenital	Minority of families
Choanal atresia	HP:0000453	Physical manifestation	Congenital	Expanded / ABERS
Pierre-Robin sequence	HP:0000201	Clinical sign	Congenital	Expanded spectrum
Coloboma	HP:0000589	Physical manifestation	Congenital	ABERS
Sensorineural hearing impairment	HP:0000407	Laboratory/ clinical	Congenital/ childhood	ABERS

Severity is variable (mild isolated ankyloglossia → severe syndromic ABERS). The malformations are **congenital and stable** (non-progressive) once present. **Quality-of-life impact** derives from speech (velopharyngeal insufficiency, hypernasality, compensatory misarticulations), feeding, hearing, and dental function; oronasal fistula and reduced speech intelligibility are the strongest predictors of poor VPI-related QOL PMID: 41520495.

4. Genetic / Molecular Information

Causal gene: TBX22 (HGNC:11600; gene MIM 300307), encoding a T-box transcription factor. **Variant types** causing CPX: nonsense, frameshift, splice-site, and T-box-domain missense PMID: 11559848. **Classification:** ClinVar documents ~170 Pathogenic, 13 Likely pathogenic, 161 VUS, 122 Benign/Likely benign. **Functional consequence:** loss of function — truncating alleles eliminate protein; T-box missense alleles abolish DNA binding and SUMO-1 conjugation at Lys63 PMID: 17846996. A gain-of-function hypothesis has been proposed specifically for ABERS PMID: 41630180. **Origin:** germline. **Allele frequency:** pathogenic alleles are rare; gnomAD shows strong LoF intolerance (pLI 0.99). **Epigenetics:** TBX22 exon-5 CpG hypermethylation downregulates expression in a teratogen model PMID: 31241704. **Chromosomal abnormalities:** large Xq21 deletions encompassing TBX22 can produce contiguous-gene phenotypes.

5. Environmental Information

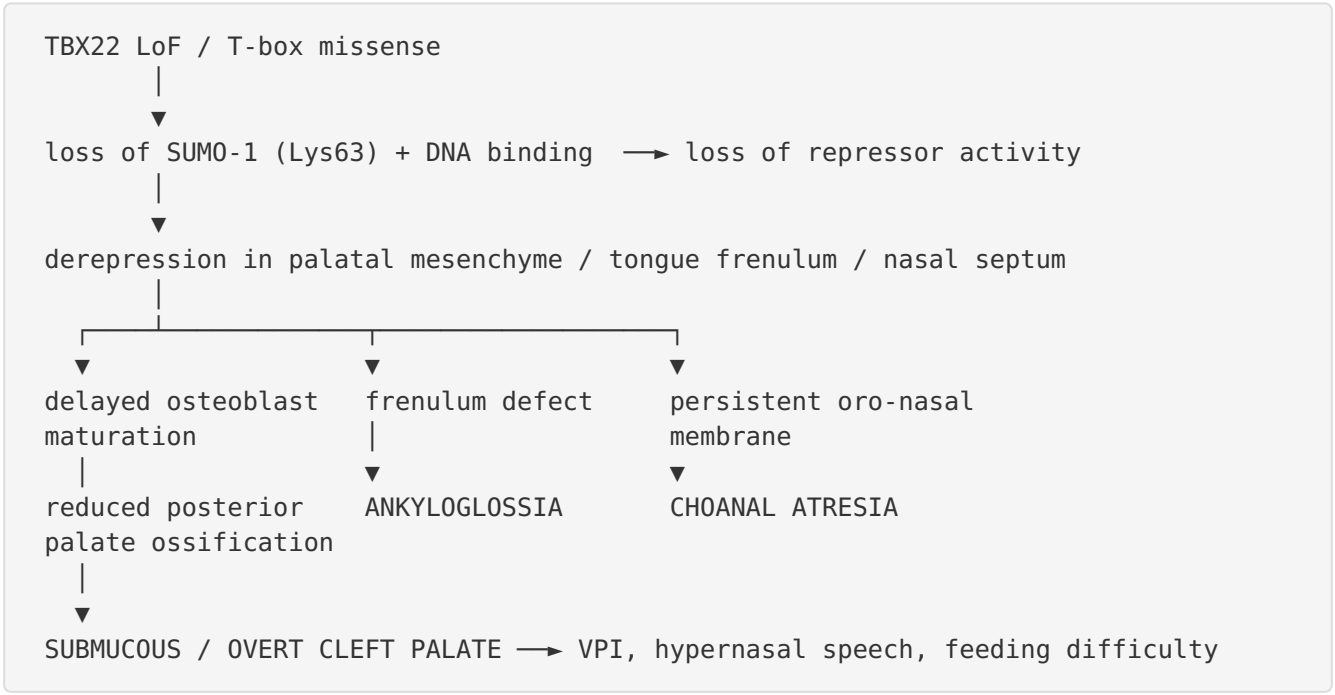
Non-genetic contributors are those of orofacial clefting generally. The best TBX22-specific datum is the **retinoic-acid (ATRA) teratogen** model, in which exposure hypermethylates TBX22 and lowers its expression during palatogenesis PMID: 31241704. Glucocorticoid/alcohol exposure models also alter Tbx22 expression during palatogenesis PMID: 19816249. No infectious agents are implicated.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A **germline loss-of-function TBX22 mutation** (truncating) or a **T-box-domain missense allele** arises on the X chromosome. → *leads to*
2. In truncating cases, **absent/nonfunctional protein**; in T-box missense cases, **loss of DNA binding at the AGGTGTGA palindrome and loss of SUMO-1 conjugation at Lys63**, which *results in* loss of repressor activity PMID: 17846996. → *leads to*
3. **Loss of TBX22 transcriptional repression** in the developing palatal mesenchyme, base of tongue, and nasal septum — tissues where TBX22 is normally expressed PMID: 12374769. (Upstream, TBX22 is normally induced by FGF8 and restricted by BMP4 PMID: 20845426.) → *results in*
4. **Delayed osteoblast maturation** after condensation of the posterior palatal mesenchyme, *leading to* **severely reduced intramembranous ossification of the posterior hard palate** PMID: 19648291. → *results in*

5. **Submucous (or overt) cleft palate** — the classic posterior notch. **Branch:** in the tongue frenulum region, loss of TBX22 *results in* **ankyloglossia**; persistence of oro-nasal membranes *results in* **choanal atresia** PMID: 19648291. → *leads to*
6. **Clinical manifestations:** feeding difficulty, velopharyngeal insufficiency, hypernasal speech, and (in syndromic ABERS) coloboma and sensorineural deafness.



Molecular pathway: TBX22 is a downstream node of **FGF8 (inducer) / BMP4 (repressor)** craniofacial signaling PMID: 20845426. **Cellular process:** osteoblast differentiation / intramembranous ossification. **Protein dysfunction:** loss of function via impaired DNA binding and loss of post-translational SUMO modification. **Epigenetic:** CpG hypermethylation represses TBX22 under teratogen exposure PMID: 31241704.

Upstream vs downstream: the mutation and loss of SUMO/DNA-binding are the most upstream events; derepression of TBX22 targets and the osteoblast-maturation delay are intermediate; the cleft, ankyloglossia, and choanal atresia are the downstream clinical endpoints.

Suggested GO terms: GO:0060021 (roof of mouth development), GO:0001649 (osteoblast differentiation), GO:0006357 (regulation of transcription by RNA Pol II), GO:0016925 (protein sumoylation), GO:0003700 (DNA-binding transcription factor activity). **Suggested CL terms:** CL:0000062 (osteoblast), CL:0000134 (mesenchymal stem cell), cranial neural-crest-derived palatal mesenchyme.

7. Anatomical Structures Affected

Primary structures: secondary palate (**UBERON:0001716** secondary palate, **UBERON:0003216** hard palate, **UBERON:0001733** soft palate), uvula (**UBERON:0010235**), tongue and lingual frenulum (**UBERON:0001723** tongue; frenulum of tongue). **Secondary involvement:** nasal septum (**UBERON:0002096**), choanae/posterior nares (**UBERON:0002220**), teeth/tooth buds (**UBERON:0001091**), middle ear (VPI-associated otitis media). In ABERS, the eye (coloboma; **UBERON:0000970**) and auditory system are involved. **Body systems:** digestive/oral, respiratory (choanae), special sense (ear, eye in ABERS). **Tissue level:** palatal **mesenchyme** (osteogenic), oral **epithelium**; **cell populations:** osteoblasts and cranial neural-crest-derived mesenchymal cells. **Subcellular:** nucleus (transcriptional repression; **GO:0005634**). **Lateralization:** cleft palate and ankyloglossia are typically **midline**.

8. Temporal Development

Onset: congenital — the lesion is established during embryonic **palatogenesis** (critical window ~weeks 6–9 in humans; E11.5–E16.5 in mouse). **Onset pattern:** structural/congenital, not acute/progressive. **Progression:** the malformation is **stable** (non-progressive); secondary functional sequelae (speech, hearing, dental, feeding) evolve through childhood and are modifiable by intervention. **Critical period:** palatal shelf elevation, mesenchymal condensation, and osteoblast maturation (the window in which TBX22 acts). **Disease duration:** lifelong structural condition, functionally improved by surgery and therapy.

9. Inheritance and Population

Inheritance: X-linked **semidominant** PMID: 11559848; PMID: 1570839. Hemizygous males more severely affected; heterozygous carrier females variably/mildly affected (X-inactivation-dependent expressivity). **Penetrance:** high in males, incomplete/variable in females. **Expressivity:** highly variable (isolated ankyloglossia → ABERS). **Sex ratio:** male-predominant for severe phenotype. **Epidemiology:** CPX is rare; TBX22 mutations account for ~**5%** of non-syndromic cleft palate. Founder kindreds were key to discovery (Icelandic, British Columbia Native, Manitoba Mennonite) PMID: 8045560. No strong consanguinity role (X-linked). **Genetic anticipation:** not applicable (not a repeat-expansion disorder). **Germline mosaicism:** possible but not specifically quantified here. (Precise CPX-specific prevalence/incidence per 100,000 are not well established in the literature.)

10. Diagnostics

Clinical diagnosis: physical examination identifies cleft palate (including submucous — bifid uvula, zona pellucida, palpable notch) and ankyloglossia; nasendoscopy and speech assessment evaluate velopharyngeal function. **Genetic testing:** targeted **single-gene TBX22 sequencing** (Sanger/NGS) is the primary confirmatory test; TBX22 is included on **cleft/orofacial and craniofacial gene panels**; **chromosomal microarray** detects Xq21 deletions; WES/WGS are used when the panel is negative or the phenotype is syndromic PMID: 30462376. **Differential diagnosis:** non-syndromic cleft palate, Van der Woude syndrome, Pierre-Robin sequence, other syndromic clefts, CHARGE syndrome (vs ABERS), and autosomal-dominant ankyloglossia with distinct loci PMID: 29704302. **Screening:** carrier testing in affected families; cascade testing; prenatal/preimplantation testing where a familial variant is known.

11. Outcome / Prognosis

Survival: normal life expectancy in isolated CPX; the mouse model's lethality (from choanal atresia/feeding) reflects severe syndromic airway compromise rather than typical human CPX. **Morbidity:** speech (velopharyngeal insufficiency, hypernasality, compensatory misarticulations), feeding, recurrent otitis media/conductive hearing loss, and dental anomalies. **Prognostic factors for speech/QOL:** timing of palatoplasty, cleft width/subtype, oronasal fistula, and speech intelligibility; "ONF (odds ratio = 8.531) and speech intelligibility deficits (odds ratio = 8.854) were independent predictors of poor QOL" PMID: 41520495. Later repair was associated with modestly increased passive cleft speech characteristics (odds ratio 1.24 per month at 36 months) PMID: 42275034. **Recovery:** good functional outcomes with timely multidisciplinary care.

12. Treatment

There is **no gene-targeted therapy**; management is the standard multidisciplinary cleft pathway.

Intervention	Description	NCIT suggestion
Palatoplasty	Primary surgical closure of the cleft palate, ideally before ~12–15 months to optimize speech	NCIT:C51756 (palatoplasty)
Furlow double-opposing Z-plasty	Associated with lower fistula and VPI rates vs alternatives	—
Frenotomy/ frenuloplasty	Release of the lingual frenulum for ankyloglossia	—
Secondary speech surgery	Pharyngoplasty / conversion Furlow for persistent VPI	—
Speech therapy	For resonance/articulation disorders	NCIT:C15415
Audiology / ENT	Grommets for otitis media, hearing support	—
Orthodontics/dental	For hypodontia and dental anomalies	—
Multimodal analgesia	Suprazygomatic maxillary nerve blocks + acetaminophen/NSAIDs (opioid-sparing)	—

Evidence: the Furlow palatoplasty "was associated with significantly lower fistula and velopharyngeal insufficiency (VPI) rates," and nerve blocks provide "robust opioid-sparing effects" PMID: 42235642. Current evidence "does not support a universally optimal timing" with anatomical factors (cleft width/subtype) more predictive than timing alone PMID: 42403778. **Pharmacogenomics / experimental / gene / cell / RNA / immunotherapies:** none established or in trials for CPX specifically.

13. Prevention

Primary prevention: periconceptional folate and avoidance of teratogens (alcohol, retinoids, smoking) for orofacial clefts generally; retinoid avoidance is mechanistically relevant given TBX22 methylation effects PMID: 31241704. **Secondary prevention:** prenatal ultrasound can detect some clefts; early postnatal examination for submucous cleft/ankyloglossia. **Genetic screening:** carrier and cascade testing in affected X-linked families; prenatal and preimplantation genetic testing where the familial TBX22 variant is known. **Counseling:** X-linked recurrence-risk counseling (an affected male transmits the allele to all daughters as carriers and no sons; a carrier female transmits to 50% of children) is central. **Tertiary prevention:** timely palatoplasty, ONF prevention, speech therapy, and audiologic surveillance prevent complications.

14. Other Species / Natural Disease

Model organism ortholog: mouse *Tbx22* (conserved T-box transcription factor; principal mouse ortholog of human TBX22). The FGF8/BMP4 regulatory relationship is conserved between **chick and mouse** PMID: 20845426. No naturally occurring companion-animal TBX22 disease was identified in this investigation (OMIA not confirmed here). Evolutionary conservation of the palatal-ossification mechanism is supported by the faithful mouse phenotype.

15. Model Organisms

Model	Type	Key phenotypes	Recapitulation
***Tbx22*-null mouse**	Mammalian knockout	Submucous cleft palate, ankyloglossia, choanal atresia, ~50% postnatal lethality	**High** — mirrors human CPX; defines osteoblast-maturation mechanism PMID: 19648291
ATRA-induced mouse cleft	Induced/ teratogen	Cleft palate with TBX22 exon-5 hypermethylation	GxE/epigenetic model PMID: 31241704
Glucocorticoid/ alcohol C57BL/6N	Induced	Cleft palate with altered Tbx22 expression	Environmental model PMID: 19816249
Chick (expression/ regulation)	Avian	FGF8/BMP4 regulation of Tbx22	Regulatory conservation PMID: 20845426

Model limitation: the mouse null produces predominantly *submucous* cleft (overt clefts are a minority), and its ~50% lethality is more severe than isolated human CPX; it does not capture the full human allelic/missense spectrum. **Resources:** MGI (mouse *Tbx22*).

Mechanistic Model / Interpretation

The CPX story is unusually complete for a Mendelian craniofacial disorder because the genetics, biochemistry, developmental biology, and a faithful animal model all converge. The gene (TBX22) was mapped by linkage and identified by mutation analysis; the biochemistry (SUMO-1/Lys63-dependent repression of an AGGTGTGA target) explains **why missense**

mutations clustered in the T-box domain are pathogenic while C-terminal missense are not; the mouse model shows **where and how** the lesion forms (posterior hard palate, via delayed osteoblast maturation rather than shelf-closure failure); and the expression pattern explains the **associated features** (frenulum → ankyloglossia; nasal septum/choanae → atresia). The population-genetic signature (pLI 0.99, LOEUF 0.48, but unconstrained missense) independently confirms a **haploinsufficiency/loss-of-function** architecture. The FGF8↑/BMP4↓ regulatory logic places TBX22 within the broader craniofacial signaling network, and the retinoid-methylation data tie the same gene to **environmentally modulated, multifactorial** clefting — a bridge between the rare Mendelian and common non-syndromic forms.

The one genuine mechanistic tension is **ABERS**: a 2026 case series proposes that gain-of-function alleles, not loss of function, may drive the severe CHARGE-like phenotype PMID: 41630180, which would mean the TBX22 allelic series is not purely a simple dosage gradient. This remains a hypothesis.

Evidence Base

PMID	Role	Contribution
11559848	Landmark	Identifies TBX22 as causal; defines loss-of-function mechanism
8045560	Linkage	Maps CPX to Xq21.1
1570839	Linkage	Carrier-female manifestation; semidominant inheritance
17846996	Mechanism	SUMO-1 / DNA-binding / repression; unifying missense mechanism
21248356	Genotype–phenotype	T-box vs C-terminal missense; expands spectrum to CLP/hypodontia
19648291	Model	Tbx22-null mouse; osteoblast-maturation mechanism
12374769	Expression	TBX22 in palate/frenulum/septum explains phenotype
22784330	Spectrum	TBX22 underlies Abruzzo-Erickson syndrome
41630180	Spectrum	ABERS case series; gain-of-function hypothesis
20845426	Regulation	FGF8 induces, BMP4 represses Tbx22
30462376	Phenotype	WES; novel signs (choanal atresia, Pierre-Robin, overgrowths)
31241704	GxE/epigenetics	Retinoid-induced TBX22 hypermethylation
25918826	Common variants	Promoter SNPs modify NSCLP risk in women
41520495	Outcomes	Predictors of VPI-related QOL
42235642	Treatment	Furlow palatoplasty, analgesia evidence
42403778	Treatment	Timing vs anatomy in outcomes
42275034	Treatment	Palatoplasty timing and speech
29704302	Differential	Autosomal-dominant ankyloglossia (genetic heterogeneity)
19816249	Model	Glucocorticoid/alcohol-induced cleft, Tbx22 expression

Limitations and Knowledge Gaps

- **CPX-specific epidemiology** (prevalence/incidence per 100,000) is not well quantified; most figures are for clefting broadly.
- **ABERS mechanism** (loss- vs gain-of-function) is unresolved PMID: 41630180.
- **Penetrance/expressivity in carrier females** is not numerically defined; X-inactivation skewing is invoked but not measured here.
- **Modifier genes** for CPX severity are not identified.
- **No human omics** (transcriptomic/proteomic) datasets specific to CPX palate were located; mechanistic detail rests on mouse and in vitro work.
- **Digital anomalies** (clinodactyly/syndactyly) are reported by Orphanet but lack a strong primary-literature frequency.
- **ClinVar Pathogenic count** is inflated by Xq21 CNVs; the intragenic point-mutation burden specific to CPX is smaller.

Proposed Follow-up Experiments / Actions

1. **Resolve the ABERS mechanism:** functional assays (DNA binding, SUMOylation, repression) on ABERS-specific splice/missense alleles to test loss- vs gain-of-function.
2. **Carrier-female study:** correlate X-inactivation ratios with phenotype severity in heterozygotes to quantify expressivity.
3. **Single-cell/spatial transcriptomics** of developing palatal mesenchyme in Tbx22-null vs WT mice to pinpoint the osteoblast-maturation program (direct TBX22 targets).
4. **ClinVar curation:** separate intragenic CPX point mutations from Xq21 CNVs to produce a clean CPX variant catalog; re-classify VUS with the SUMO/T-box functional data.
5. **GxE validation:** test whether TBX22 promoter-SNP carriers have altered methylation responses to retinoid/folate status in human cohorts.
6. **Natural-history/QOL registry** for genetically confirmed CPX to define CPX-specific speech, hearing, and dental outcomes distinct from non-syndromic cleft palate.